Chapter 7. Nonlinear Regression

Lecture 6-4 (polynomials, step functions, basis functions, regression splines, smoothing splines, local regression, GAM)

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1 Basis-Function Regression

1.1 Polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i.$$

For large enough d this produces an extremely nonlinear curve, yet the parameters are estimated by OLS: create $X_1 = X$, $X_2 = X^2$, ... and treat it as multiple linear regression. We are rarely interested in the coefficients; we care about the fitted function

$$\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \hat{\beta}_2 x_0^2 + \hat{\beta}_3 x_0^3 + \hat{\beta}_4 x_0^4,$$

with pointwise interval $\hat{f}(x_0) \pm t_{1-\alpha/2,\mathrm{df}}\,\mathrm{SE}[\hat{f}(x_0)]$, where $\mathrm{SE}[\hat{f}(x_0)]^2 = \ell_0^\top \widehat{\mathrm{Var}}(\hat{\beta})\ell_0$ with $\ell_0 = (1, x_0, \ldots, x_0^4)^\top$. Fix d at a low value (rarely above 3 or 4) or choose it by CV. Example: a degree-4 polynomial of wage on age, with 95% bands; on the Auto data, $d = 1$ underfits mpg on horsepower, $d = 2$ fits well, $d = 14$ is wild.

Limitation. Polynomials are global: the fit in one region is affected by data far away, the tails become erratic for large d, and a polynomial may fail to recover a nonlinear pattern even when d is huge.

1.2 Step functions

Cut the range of X at $c_1 < \cdots < c_K$:

$$C_0(X) = \mathbb{I}(X < c_1),\ C_1(X) = \mathbb{I}(c_1 \le X < c_2),\ \ldots,\ C_K(X) = \mathbb{I}(c_K \le X),$$

so $C_0 + \cdots + C_K = 1$, and fit $y_i = \beta_0 + \beta_1 C_1(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$ (C_0 is dropped because of the intercept). β_0 is the mean for $X < c_1$ and β_k the difference of bin k from it. With bins < 35, 35–50, 50–65, ≥ 65 for wage on age, the first bin misses the increasing trend; still, step functions are popular in biostatistics and epidemiology (e.g. five-year age groups). The result depends on the breakpoints.

1.3 Basis functions

Both are special cases of

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

with fixed, known functions b_k: $b_j(x) = x^j$ for polynomials and $b_j(x) = \mathbb{I}(c_j \le x < c_{j+1})$ for step functions. Everything of linear regression (least squares, SEs, F-tests) applies. Other bases: Fourier series, wavelets (e.g. for image denoising), and the splines below.

2 Regression Splines

2.1 Piecewise polynomials

Fit separate low-degree polynomials over regions separated by **knots**; with one knot c,

$$y_i = \begin{cases} \beta_{01} + \beta_{11} x_i + \beta_{21} x_i^2 + \beta_{31} x_i^3 + \varepsilon_i, & x_i < c, \\ \beta_{02} + \beta_{12} x_i + \beta_{22} x_i^2 + \beta_{32} x_i^3 + \varepsilon_i, & x_i \ge c. \end{cases}$$

More knots give more flexibility. Unconstrained pieces jump at the knot, so we add constraints (continuity, continuity of derivatives). Splines have the "maximum" amount of continuity.

2.2 Linear splines

Definition: linear spline

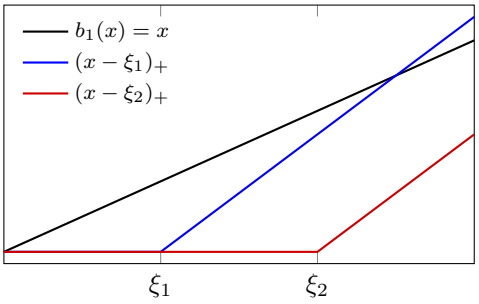
A linear spline with knots $\xi_1 < \cdots < \xi_K$ is piecewise linear and continuous at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+1} b_{K+1}(x_i) + \varepsilon_i, \qquad b_1(x) = x, \quad b_{k+1}(x) = (x - \xi_k)_+,$$

with $(x - \xi)_+ = x - \xi$ if $x > \xi$ and 0 otherwise.

Each truncated term $(x - \xi_k)_+$ adds a change of slope β_{k+1} at ξ_k while keeping continuity.

(a) basis functions



(b) linear spline

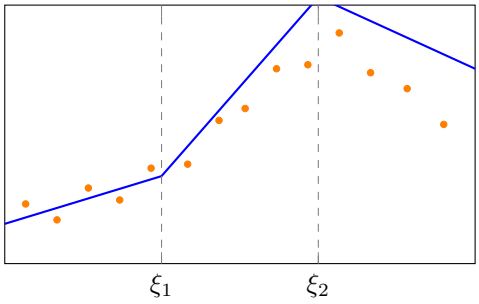


Figure 1: (a) The truncated power basis of a linear spline with knots ξ_1, ξ_2. (b) A fit $\beta_0 + \beta_1 x + \beta_2(x - \xi_1)_+ + \beta_3(x - \xi_2)_+$: continuous, with slope changes β_2 at ξ_1 and β_3 at ξ_2.

2.3 Cubic splines

Definition: cubic spline and truncated power basis

A cubic spline with knots $\xi_1, \ldots, \xi_K$ is piecewise cubic with continuous derivatives up to order 2 at each knot:

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \varepsilon_i,$$

$$b_1(x) = x, \; b_2(x) = x^2, \; b_3(x) = x^3, \; b_{k+3}(x) = (x - \xi_k)_+^3, \; k = 1, \ldots, K.$$

Theorem: degrees of freedom of a cubic spline

The space of cubic splines with K knots has dimension $K + 4$, and the truncated power basis (with the constant) is a basis.

Proof [Supplement]. $K + 1$ unconstrained cubic pieces have $4(K + 1)$ parameters; continuity of the function, first and second derivatives at each of the K knots imposes $3K$ linear constraints, leaving $4(K + 1) - 3K = K + 4$. Each $(x - \xi_k)_+^3$ is twice continuously differentiable (its first two derivatives vanish at ξ_k) and its third derivative jumps by 6 at ξ_k, so the $K+4$ functions are linearly independent splines. ■

With continuity alone the fit can have a kink; continuity of the first and second derivatives makes it visually smooth.

2.4 Natural cubic splines

Splines can have high variance at the outer range of the predictor (few data, wide confidence bands). A **natural cubic spline** additionally requires the function to be *linear* beyond the boundary knots.

This adds 4 constraints (second and third derivatives zero at both ends), so a natural cubic spline with K knots has K degrees of freedom (including the intercept), and its confidence bands are narrower at the boundaries.

2.5 Number and locations of knots

Decide the number K (or the degrees of freedom) and place knots at quantiles of the observed X (or let software do it), so that each region has a similar amount of data. Choose K by cross-validation. On the Wage data, a degree-15 polynomial is wild at the boundaries, whereas a natural cubic spline with 15 df is well behaved: splines add flexibility through knots while keeping the degree fixed.

3 Smoothing Splines

Motivation. Instead of choosing knots, let every data point be a knot and control wiggliness by a roughness penalty. The minimizer turns out to be a natural cubic spline, and the penalty weight λ plays the role of the number of knots.

Definition: smoothing spline

Among all twice-differentiable functions g,

$$\min_g \underbrace{\sum_{i=1}^{n}(y_i - g(x_i))^2}_{\text{RSS}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness penalty}} .$$

The first term makes g match the data; the second controls how wiggly g is (g'' measures curvature), modulated by $\lambda \geq 0$. The smaller λ, the more wiggly the function, eventually interpolating the y_i at $\lambda = 0$; as $\lambda \to \infty$, g becomes linear (any linear function has zero penalty). λ can be chosen by K-fold CV.

Theorem: the minimizer is a natural cubic spline

The minimizer $\hat{g}_\lambda$ is a natural cubic spline with knots at the unique values of $x_1, \ldots, x_n$ — a shrunken version of the interpolating natural spline. Consequently $\hat{g}_\lambda = (\hat{g}(x_1), \ldots, \hat{g}(x_n))^\top = S_\lambda y$ for an $n \times n$ smoother matrix S_λ.

Idea [Supplement]. Given any candidate g, let $\tilde{g}$ be the natural cubic spline interpolating the values $g(x_i)$. Then RSS is unchanged, and an integration-by-parts argument shows $\int \tilde{g}''^2 \leq \int g''^2$, with equality only if $g = \tilde{g}$. So the search can be restricted to natural cubic splines with knots at the x_i, which form an n-dimensional linear space $g(x) = \sum_j N_j(x)\theta_j$. The problem becomes $\min_\theta \|y - N\theta\|^2 + \lambda\theta^\top \Omega_N \theta$ with $(\Omega_N)_{jk} = \int N_j'' N_k''$, solved by $\hat{\theta} = (N^\top N + \lambda\Omega_N)^{-1} N^\top y$, a generalized ridge regression. Hence $S_\lambda = N(N^\top N + \lambda\Omega_N)^{-1} N^\top$. ■

The **effective degrees of freedom** $\mathrm{df}_\lambda = \mathrm{tr}(S_\lambda)$ decreases from n ($\lambda = 0$) to 2 ($\lambda \to \infty$) and is a convenient way to specify λ. Because $\hat{g}$ is a linear smoother, LOOCV costs one fit: $\mathrm{CV}(\lambda) = \frac{1}{n}\sum_i \{(y_i - \hat{g}_\lambda(x_i))/(1 - \{S_\lambda\}_{ii})\}^2$ (Ch. 3). On the Auto and Wage data, a smoothing spline with about 6.8 effective df (chosen by LOOCV) is nearly indistinguishable from a 16-df fit.

4 Local Regression

With a sliding weight function, fit separate (weighted) linear fits over the range of X.

Algorithm: local regression at a target point x_0

1. Gather the fraction $s = k/n$ of training points closest to x_0 (the **span**).
2. Assign weights $K_{i0} = K(x_i, x_0)$: zero outside the neighborhood, largest for the closest points (e.g. tricube weights).
3. Fit a weighted least-squares regression $\min_{\beta_0,\beta_1} \sum_i K_{i0}(y_i - \beta_0 - \beta_1 x_i)^2$.
4. The fitted value is $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Choices: the weighting function, whether to fit a constant, linear or quadratic regression locally, and the span s, which plays the role of λ: small s gives a wiggly fit, large s a global one; choose s by CV. Local linear fits remove the boundary bias that local constant (kernel) averages have. Like KNN, local regression suffers from the curse of dimensionality for more than three or four predictors.

5 Generalized Additive Models

GAMs allow flexible nonlinearities in several variables while retaining the additive structure of linear models:

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \varepsilon_i.$$

A GAM can be fitted with natural splines (then it is one big least-squares regression on the combined basis), with smoothing splines or local regression, or mixing terms — some linear, some nonlinear, some step functions.

Algorithm: backfitting (for smoothing-spline or local-regression terms)

Initialize $\hat{\beta}_0 = \bar{y}$, $\hat{f}_j = 0$. Repeat until convergence: for $j = 1, \ldots, p$, smooth the partial residuals $r_i = y_i - \hat{\beta}_0 - \sum_{k \neq j} \hat{f}_k(x_{ik})$ against x_{ij} to update $\hat{f}_j$, then center $\hat{f}_j$ to have mean zero.

Each f_j is updated with the others held fixed, which is possible because the model is additive.

Example: Wage data

$\texttt{wage} = \beta_0 + f_1(\texttt{year}) + f_2(\texttt{age}) + f_3(\texttt{education}) + \varepsilon$, with natural splines for year and age and a separate constant for each education level. Holding the others fixed, wage rises slightly with year, rises and then falls with age, and increases with education.

Pros. A nonlinear f_j for each X_j without manually trying many transformations; the additive structure still allows examining the effect of each X_j with the others fixed; potentially more accurate predictions. Effective degrees of freedom summarize each f_j. **Cons.** Additivity can miss important interactions (they can be added by hand, e.g. $f_{jk}(X_j, X_k)$ via a two-dimensional smoother).

Summary

- Basis-function regression (polynomials, step functions, splines) stays within linear least squares.
- Splines are piecewise polynomials joined smoothly at knots: linear splines $(x - \xi)_+$, cubic splines $(x - \xi)_+^3$ with $K + 4$ df, natural cubic splines linear beyond the boundary (K df).

- Smoothing splines $\min \mathrm{RSS} + \lambda \int g''^2$ yield natural cubic splines with knots at the data; $\mathrm{df}_\lambda = \mathrm{tr} S_\lambda$.
- Local regression fits weighted least squares in a sliding window (span s).
- GAMs combine univariate nonlinear functions additively (backfitting); interactions must be added explicitly.

6 Selected exercises

Motivation. ISLR Exercises 7.1 and 7.2 check the truncated power basis and the role of the penalty order; the last item completes the integration-by-parts argument behind the smoothing-spline theorem.

Exercise 7.1 (The truncated power basis gives a cubic spline; ISLR Ex. 7.1)

Let $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x-\xi)_+^3$.
(a) Find cubics $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$ and $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$ with $f = f_1$ on $x \le \xi$ and $f = f_2$ on $x > \xi$.
(b) Show that $f_1(\xi) = f_2(\xi)$, $f_1'(\xi) = f_2'(\xi)$ and $f_1''(\xi) = f_2''(\xi)$, so f is a cubic spline with knot ξ.

Solution.

(a) For $x \le \xi$ the truncated term vanishes: $a_1 = \beta_0$, $b_1 = \beta_1$, $c_1 = \beta_2$, $d_1 = \beta_3$. For $x > \xi$, $(x-\xi)^3 = x^3 - 3\xi x^2 + 3\xi^2 x - \xi^3$, so

$$a_2 = \beta_0 - \beta_4 \xi^3, \quad b_2 = \beta_1 + 3\beta_4 \xi^2, \quad c_2 = \beta_2 - 3\beta_4 \xi, \quad d_2 = \beta_3 + \beta_4.$$

(b) As polynomials, $f_2 - f_1 = \beta_4 (x-\xi)^3$, with derivatives $3\beta_4(x-\xi)^2$ and $6\beta_4(x-\xi)$. All three vanish at $x = \xi$, which gives the three equalities. The third derivatives differ by $6\beta_4$, so f''' jumps at ξ unless $\beta_4 = 0$: β_4 is exactly the size of the jump, divided by 6. ■

Exercise 7.2 (Which penalty, which limit; ISLR Ex. 7.2)

Let $x_1 < \cdots < x_n$ be distinct points in $[a, b]$, $n \ge 3$, and

$$\hat{g} = \arg\min_g \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int_a^b \left(g^{(m)}(x)\right)^2 dx$$

over functions with continuous mth derivative ($g^{(0)} = g$). Interpreting $\lambda = \infty$ as the constraint that the penalty vanishes, describe $\hat{g}$ for $\lambda = \infty$ and $m = 0, 1, 2, 3$, and for $\lambda = 0$, $m = 3$.

Solution.

Step 1. A continuous nonnegative function with zero integral vanishes identically, so $\int_a^b (g^{(m)})^2 = 0$ iff $g^{(m)} \equiv 0$ on $[a, b]$. For $m = 0$ this means $g \equiv 0$; for $m \ge 1$, integrating m times, it means that g is a polynomial of degree at most $m - 1$.

Step 2. With $\lambda = \infty$, $\hat{g}$ minimizes RSS over that class:

(a) $m = 0$: $\hat{g} \equiv 0$.

(b) $m = 1$: constants; $\sum_i (y_i - c)^2$ is minimized at $c = \bar{y}$, so $\hat{g} \equiv \bar{y}$.

(c) $m = 2$: lines; $\hat{g}$ is the least-squares line.

(d) $m = 3$: quadratics; $\hat{g}$ is the least-squares quadratic, unique because three or more distinct x_i make the design $(1, x_i, x_i^2)$ of full column rank.

Step 3. With $\lambda = 0$ and $m = 3$, every smooth g with $g(x_i) = y_i$ for all i attains RSS $= 0$, the smallest possible value (for instance the interpolating polynomial of degree $n-1$). So $\hat{g}$ interpolates the data, and it is not unique. Large λ pulls the fit to the null space of the penalty (polynomials of degree $< m$); small λ lets it chase the data. ■

Exercise 7.3 (The natural cubic spline has least curvature)

Let $a < x_1 < \cdots < x_n < b$ with $n \geq 2$, and let $\tilde{g}$ be a natural cubic spline with knots $x_1, \ldots, x_n$ (linear on $[a, x_1]$ and $[x_n, b]$). Let g be any twice continuously differentiable function on $[a, b]$ with $g(x_i) = \tilde{g}(x_i)$ for all i. Show that $\int_a^b g''^2 \geq \int_a^b \tilde{g}''^2$, with equality iff $g = \tilde{g}$. Deduce that the smoothing-spline minimizer is a natural cubic spline with knots at the x_i.

Solution [Supplement]. Let $h = g - \tilde{g}$; it is twice continuously differentiable and $h(x_i) = 0$ for all i.

Step 1. $\tilde{g}''$ is continuous and piecewise linear, hence absolutely continuous with derivative $\tilde{g}'''$ almost everywhere, where $\tilde{g}'''$ equals a constant c_j on (x_j, x_{j+1}) and 0 on (a, x_1) and (x_n, b). Integration by parts gives

$$\int_a^b \tilde{g}'' h'' = \left[\tilde{g}'' h'\right]_a^b - \int_a^b \tilde{g}''' h'.$$

The boundary term vanishes because $\tilde{g}$ is linear near a and b, so $\tilde{g}''(a) = \tilde{g}''(b) = 0$.

Step 2. $\int_a^b \tilde{g}''' h' = \sum_{j=1}^{n-1} c_j \int_{x_j}^{x_{j+1}} h' = \sum_{j=1}^{n-1} c_j \{h(x_{j+1}) - h(x_j)\} = 0$. Hence $\int \tilde{g}'' h'' = 0$.

Step 3. Therefore

$$\int_a^b g''^2 = \int_a^b (\tilde{g}'' + h'')^2 = \int_a^b \tilde{g}''^2 + \int_a^b h''^2 \;\geq\; \int_a^b \tilde{g}''^2.$$

Step 4. Equality forces $\int h''^2 = 0$, so $h'' \equiv 0$ (continuity) and h is linear. A linear function vanishing at $n \geq 2$ points is 0, so $g = \tilde{g}$.

Step 5 (consequence). Given any candidate g for the smoothing-spline problem, let $\tilde{g}$ be the natural cubic spline interpolating the values $g(x_i)$ (it exists, as the chapter uses). The RSS is the same and, by Steps 3–4, the penalty strictly decreases unless $g = \tilde{g}$. Hence every minimizer is a natural cubic spline with knots at the x_i. ■